

Copula-Based Anomaly Detection in Spatio-Temporal Seismic Data

1 Introduction

Seismic data, inherently spatio-temporal, require methods that account for dependencies among signals from multiple stations. Traditional univariate approaches for anomaly detection pool p-values but fail to consider spatial and temporal dependencies. Copulas, which decouple marginal distributions from dependency structures, offer a robust framework for addressing these challenges. This proposal outlines a methodology integrating spatio-temporal modeling, time series alignment, and copula-based pooling to improve anomaly detection accuracy.

2 Proposed Methodology

2.1 Spatio-Temporal VARIMA-X Model

To capture seismic dynamics, we model the observations $\mathbf{y}_t = [y_{1,t}, \dots, y_{k,t}]^\top$ from k stations using a VARIMA-X model:

$$(1 - B)^d \mathbf{y}_t = \sum_{i=1}^p \Phi_i (1 - B)^d \mathbf{y}_{t-i} + \sum_{j=1}^q \Theta_j \mathbf{w}_{t-j} + \Gamma \mathbf{x}_t + \mathbf{w}_t, \quad (1)$$

where B is the backshift operator, d the differencing order, Φ_i and Θ_j the autoregressive and moving average coefficients, $\mathbf{x}_t$ the exogenous covariates, and $\mathbf{w}_t \sim N(0, Q)$ the error term.

2.2 Time Series Alignment with DTW

Dynamic Time Warping (DTW) aligns time series by minimizing cumulative distance under constraints:

$$W = \arg \min_W \sum_{(i,j) \in W} |a_i - b_j|, \quad (2)$$

where A and B are the time series to be aligned. Alignment ensures residuals reflect true seismic anomalies.

2.3 Copula-Based p-Value Pooling

Using Sklar’s Theorem, the joint distribution of p-values $\mathbf{p}_t = (p_{1,t}, \dots, p_{k,t})$ is modeled via a copula C :

$$C(u_1, \dots, u_k) = F(F_1^{-1}(u_1), \dots, F_k^{-1}(u_k)), \quad (3)$$

where F is the joint distribution and F_i the marginals. The combined p-value is computed as:

$$p_{\text{combined},t} = 1 - C(p_{1,t}, \dots, p_{k,t}). \quad (4)$$

Anomalies are flagged if $p_{\text{combined},t} < \alpha$.

3 Theoretical Validation

3.1 Asymptotic Normality of Residuals

Under regular conditions, residuals from the VARIMA-X model converge to normality, ensuring valid hypothesis testing.

3.2 Copula-Based Dependency Modeling

Copulas effectively capture dependencies among p-values, enhancing sensitivity to anomalies while controlling false positives.

4 Simulation Study

To determine the optimal order for the simulation, we applied the Auto-ARIMA function from the pmdarima package to the AFI station (closest station to Fuji) dataset. This method automatically selects the best-fitting ARIMA(p, d, q) parameters by minimizing an information criterion, such as AIC or BIC.

The optimal (p, d, q) order was used to fit an ARIMA model to the data, and the corresponding coefficients were extracted. Using these same parameters, we simulated data from an ARIMAX model, generating a multivariate time series dataset with two explanatory variables by the following algorithm.

4.1 ARIMAX Simulation Function

This function simulates an ARIMAX time series with exogenous variables. It generates a multivariate time series dataset based on given autoregressive (AR), moving average (MA), and exogenous variable parameters.

4.1.1 Algorithm Explanation

1. Initialize Variables

- m : Number of variables.

- p : AR order (length of ϕ).
- q : MA order (length of θ).
- Generate multivariate normal noise ω with given mean and covariance.

2. Create Data Storage

- Initializes a simulated data matrix of shape (n, m) .
- Sets the first few values (based on p or q) from the initial state.

3. Generate Time Series Data

- **AR term:** Uses past p values from the simulated data (reversed) to calculate the AR effect.
- **MA term:** Uses past q noise values (ω) for the MA effect.
- **Compute each time step** using the formula:

$$X_t = c + \sum_{i=1}^p \phi_i X_{t-i} + \sum_{j=1}^q \theta_j \omega_{t-j} + \omega_t \quad (5)$$

4. Add Exogenous Variables

- Uses the second and third columns of the simulated data as explanatory variables.
- The main time series (first column) is adjusted by adding a weighted sum of the exogenous variables using β .

5. Return the Simulated Data

4.1.2 Parameters and Output

Parameters:

- c : Constant term (intercept).
- ϕ : AR coefficients (defines how past values influence the current value).
- θ : MA coefficients (defines how past noise influences the current value).
- β : Exogenous variable coefficients (weights for explanatory variables).
- ω_{mean} : Mean vector for noise.
- ω_{Σ} : Covariance matrix for noise.
- *initial state*: Starting values for the time series.
- n : Number of time steps (length of generated data).

- *seed*: Random seed for reproducibility.

Output: A 2D NumPy array of shape (n, m) , where:

- The first column represents the main time series (influenced by AR, MA, and exogenous variables).
- The other columns are the explanatory variables.

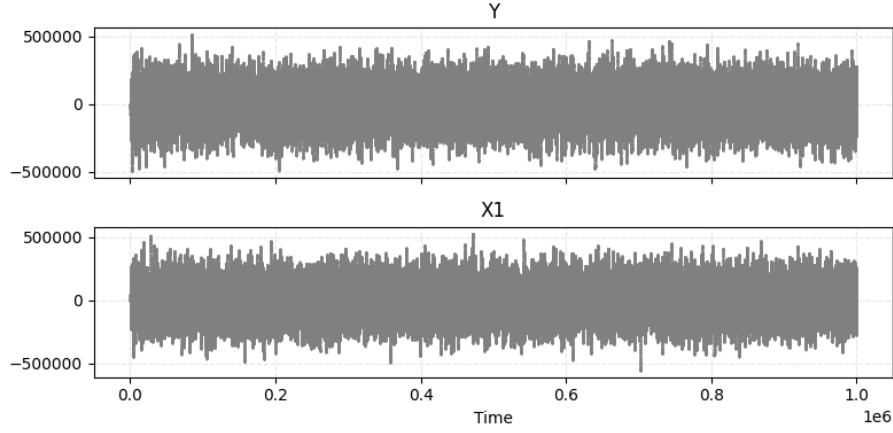


Figure 1: The output of ARIMAX Simulation Function

4.2 Earthquake Simulation Function

This function introduces earthquake anomalies into the simulated time series data. Each earthquake event starts at a random time step, gradually increases in intensity until reaching a peak, and then weakens symmetrically. The anomalies are modeled using a linear growth and decay function with added noise.

4.2.1 Algorithm Explanation

1. Initialize Variables

- `num_earthquakes`: Number of earthquake events to simulate.
- `scale`: Strength of the earthquake effect.
- `duration`: Length of each earthquake event.

2. Generate Earthquake Events

- Choose a random start time t_{start} for each earthquake.
- Compute the peak intensity time as $t_{\text{peak}} = t_{\text{start}} + \frac{\text{duration}}{2}$.

- Compute the end time as $t_{\text{end}} = t_{\text{start}} + \text{duration}$.

3. Apply Earthquake Effect

- If $t_{\text{start}} \leq t \leq t_{\text{peak}}$, the anomaly grows linearly:

$$E_t = \frac{t - t_{\text{start}}}{t_{\text{peak}} - t_{\text{start}}} \cdot S \cdot \epsilon_t \quad (6)$$

- If $t_{\text{peak}} < t \leq t_{\text{end}}$, the anomaly weakens linearly:

$$E_t = \frac{t_{\text{end}} - t}{t_{\text{end}} - t_{\text{peak}}} \cdot S \cdot \epsilon_t \quad (7)$$

4. Modify the Time Series Data

- Add the computed earthquake effect E_t to the original data for all affected variables.

4.2.2 Parameters and Output

Parameters:

- simulated_data: Original multivariate time series data.
- num_earthquakes: Number of earthquake events to simulate.
- scale: Strength of the earthquake effect.
- duration: Length of each earthquake event.
- seed: Random seed for reproducibility.

Output: A modified version of the input time series dataset, where earthquake anomalies have been introduced at random points.

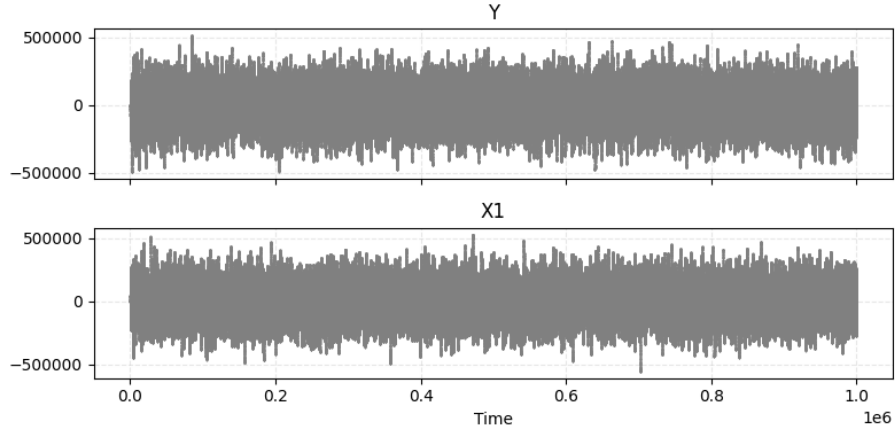


Figure 2: The output of Earthquake Simulation Function

4.3 Extracting the Residual Variance (σ^2) from ARIMA Models

We fit an **AutoRegressive Integrated Moving Average (ARIMA)** model to overlapping time windows of the earthquake dataset. The primary objective is to estimate the **residual variance** (σ^2) for each time window, which provides insights into the variability of the time series.

4.3.1 Sliding Window Segmentation

The earthquake data is divided into overlapping windows of **200 samples**, with an overlap of **20 samples** between consecutive windows. This segmentation is performed separately for each of the three time series variables.

4.3.2 ARIMA Model Fitting

For each window, an **ARIMA(4,0,0)** model is fitted. The chosen order represents:

- **4 autoregressive (AR) terms** – capturing past dependencies.
- **0 differencing (I term)** – assuming stationarity.
- **0 moving average (MA) terms** – focusing purely on the autoregressive structure.

4.3.3 Extracting σ^2 from the ARIMA Model

Once the model is fitted, the variance of the residual errors (σ^2) is extracted. This value quantifies the amount of unexplained variability after accounting for the autoregressive structure.

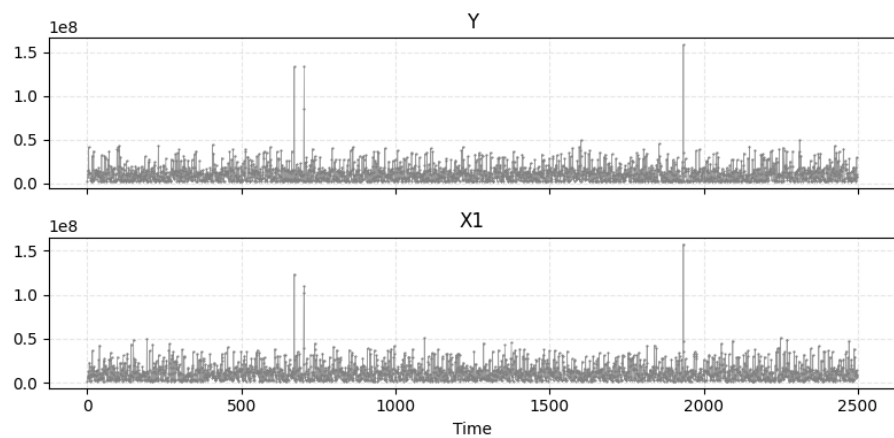


Figure 3: Sigma2 values extracted from different windows

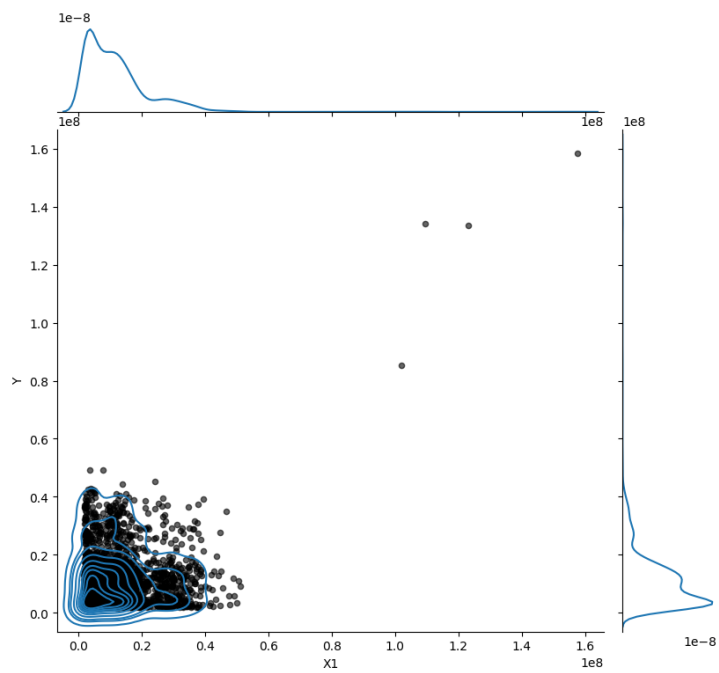


Figure 4: kernel density estimate (KDE) plot of two variables

4.4 Applied Copula

Estimate the joint cumulative distribution function (CDF) of two variables using a Gaussian copula. The method includes transforming the empirical marginals into standard normal quantiles, estimating the correlation, computing the joint CDF, and finally flagging extreme values.

We Estimate the joint CDF of two variables using a Gaussian copula. The approach involves:

1. Extracting and defining the sigma2.
2. Computing the empirical CDF to obtain uniform marginals.
3. Transforming these uniform marginals to standard normal quantiles.
4. Estimating the dependence (correlation) between the transformed variables.
5. Calculating the joint CDF using the Gaussian copula.
6. Flagging and visualizing extreme values that exceed a pre-defined threshold.

4.4.1 Step 1: Estimate the Marginals using the Empirical CDF

The empirical CDF is calculated by ranking the data. We use the `rankdata` function with the *ordinal* method so that the smallest value is assigned rank 1. Dividing by `n + 1` normalizes the ranks, resulting in values uniformly distributed on the interval $[0, 1]$.

4.4.2 Step 2: Transform the Uniform Marginals to Standard Normal Quantiles

To apply the Gaussian copula, the uniform marginals are transformed into standard normal quantiles. This transformation is performed using the inverse of the standard normal CDF, Φ^{-1} .

4.4.3 Step 3: Estimate the Dependence (Correlation) using the Transformed Data

Next, we estimate the dependence between `z_x` and `z_y` by computing their Pearson correlation coefficient. This correlation, denoted by ρ_{est} , is later used in the Gaussian copula to model the joint distribution.

4.4.4 Step 4: Compute the Joint CDF using the Gaussian Copula

The joint CDF using the Gaussian copula is defined as:

$$F(x, y) = \Phi_{\rho}(z_x, z_y)$$

where Φ_ρ is the bivariate normal CDF with correlation ρ . We define a function to compute the joint CDF for a given pair (z_1, z_2) and then calculate the joint CDF for every observation.

4.4.5 Visualization and Flagging Extreme Values

We set a significance level $\alpha = 0.05$ and flag the points where the joint CDF exceeds $1 - \alpha$. These flagged points are highlighted on the scatter plot with a red star marker. Each flagged point is also annotated with its corresponding joint CDF value.

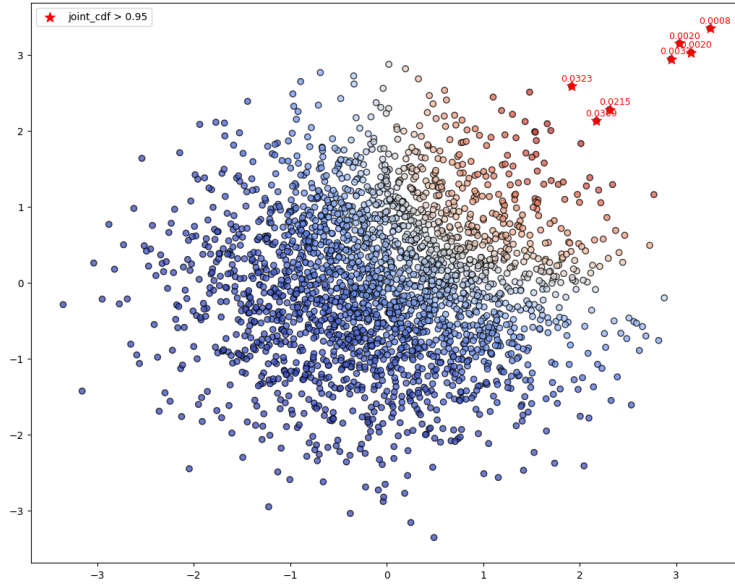


Figure 5: The flagged points with their joint CDF values

5 Case Study: Real Seismic Data

We will apply this framework to seismic data from multiple stations, highlighting its ability to detect anomalies corresponding to seismic events. DTW ensures alignment, while copulas model dependencies among station-specific p-values. The idea is to show with high confidence that pooling p-values from aligned seismic waves provides a better anomaly characterization than with one wave alone.

5.1 Source of Seismic Data

We have used the `fdsnws-datasetselect` service, which allows access to time series data for specified channels and time periods. We used the "IU" network which is Global Seismographic Network (GSN) operated by IRIS in collaboration with the USGS.

Download Link: EarthScope Data Center (IRISDMC).

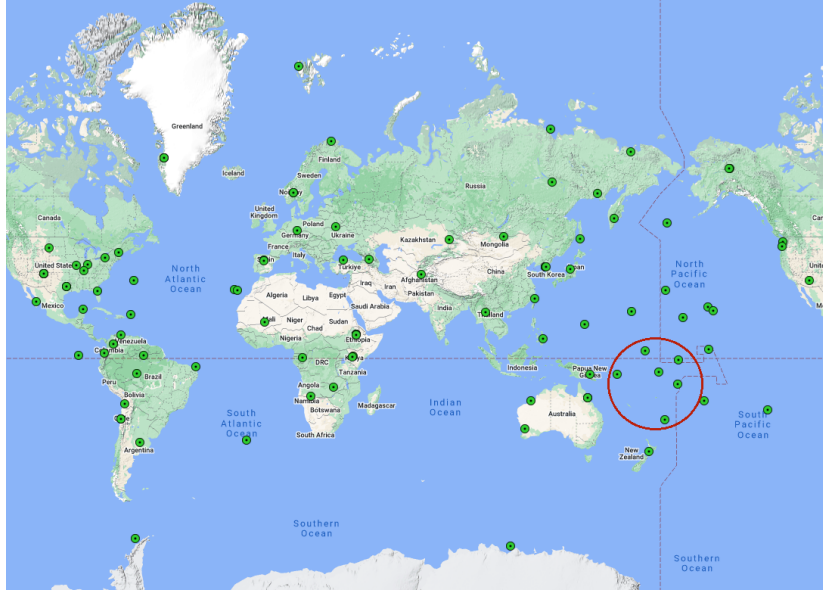


Figure 6: IU: Global Seismograph Network Stations (GSN - IRIS/USGS)

5.2 Stations

Six stations closest to Fiji were selected for the analysis (indicated by the red circle on the map). However, no data was available for the KNTN station. The downloaded data spans from 2018-08-18T12:00:00 to 2018-08-19T12:00:00, covering the period before and after the earthquake.

Station Code	Station Name	Latitude	Longitude	Start	End	Data Center(s)
TARA	Tarawa Island, Republic of Kiribati	1.3549	172.9229	2007-01-01	2599-12-31	IRISDMC
HNR	Honiara, Solomon Islands	-9.4387	159.9475	1993-10-22	2599-12-31	IRISDMC
RAO	Raoul, Kermadec Islands	-29.245	-177.929	2004-07-18	2599-12-31	IRISDMC
AFI	Afiamalu, Samoa	-13.9085	-171.7827	1993-08-24	2599-12-31	IRISDMC
FUNA	Funafuti, Tuvalu	-8.5259	179.1966	2004-01-21	2599-12-31	IRISDMC
KNTN	Kanton, Kiritibati	-2.7744	-171.7186	2007-12-04	2599-12-31	IRISDMC

Table 1: Selected Stationes close to Fiji

5.3 Exploratory Data Analysis

In the following graphs, the raw data is shown for the five stations (TARA, HNR, RAO, AFI, FUNA). The downloaded data from TARA, FUNA, and RAO stations includes two time series, which are presented in separate graphs. A significant fluctuation can be observed in the time series around 00:20 AM, corresponding to the main earthquake event. Additional fluctuations are seen post-earthquake, between 04:30 AM and 08:00 AM. Furthermore, some variations are noticeable on August 18th, particularly between 3:00 PM and 10:00 PM at all stations except FUNA.

5.4 Raw Data

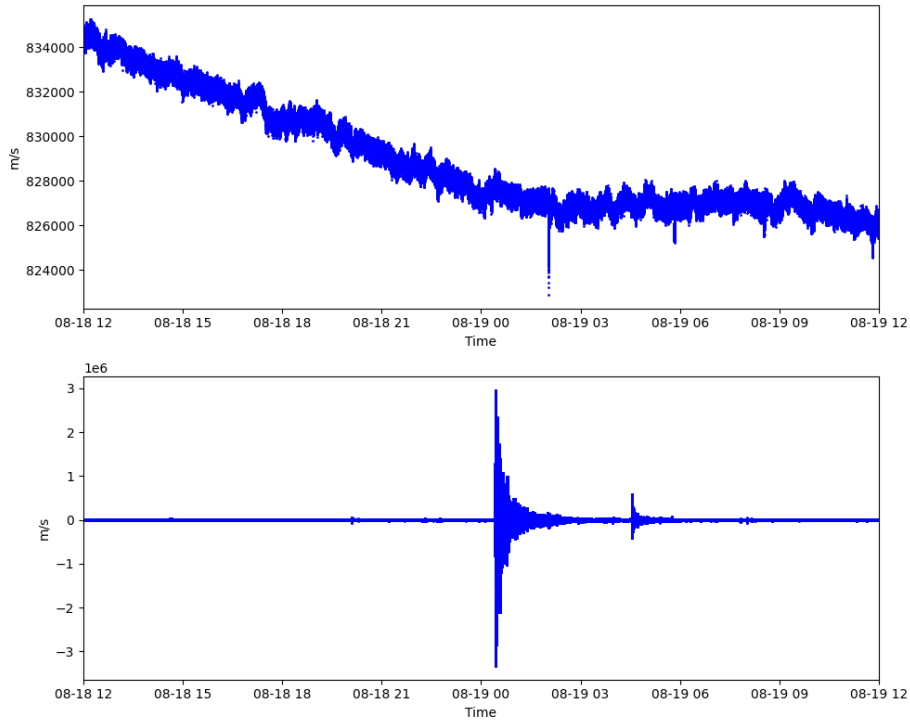


Figure 7: Raw Fiji Earthquake Time Series Data from TARA station

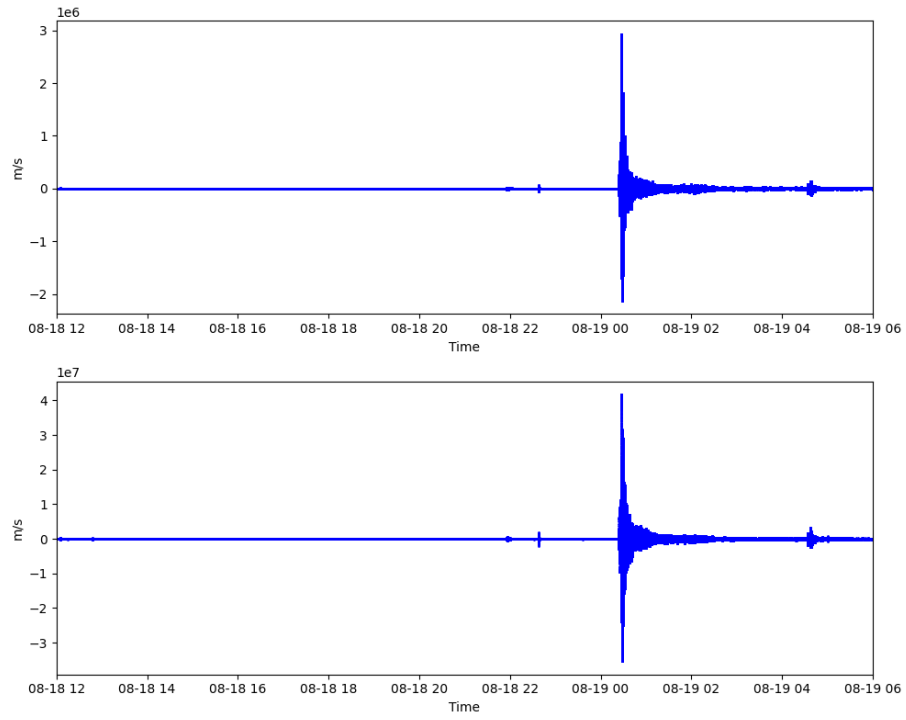


Figure 8: Raw Fiji Earthquake Time Series Data from HNR station

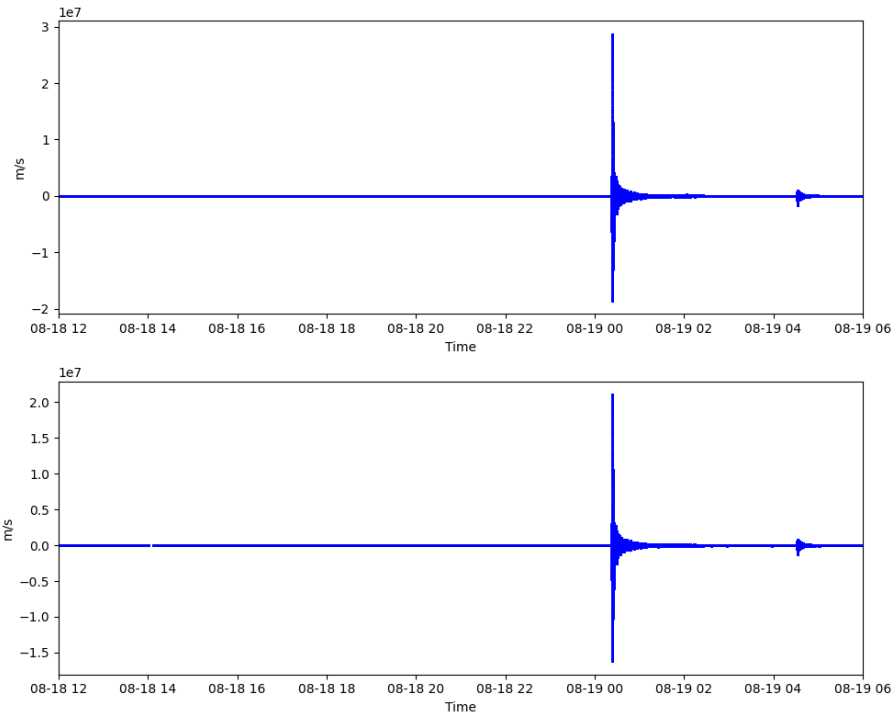


Figure 9: Raw Fiji Earthquake Time Series Data from FUNA station

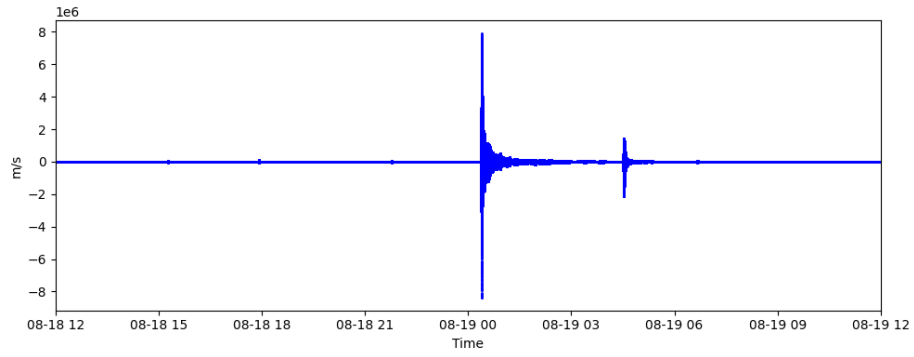


Figure 10: Raw Fiji Earthquake Time Series Data from AFI station

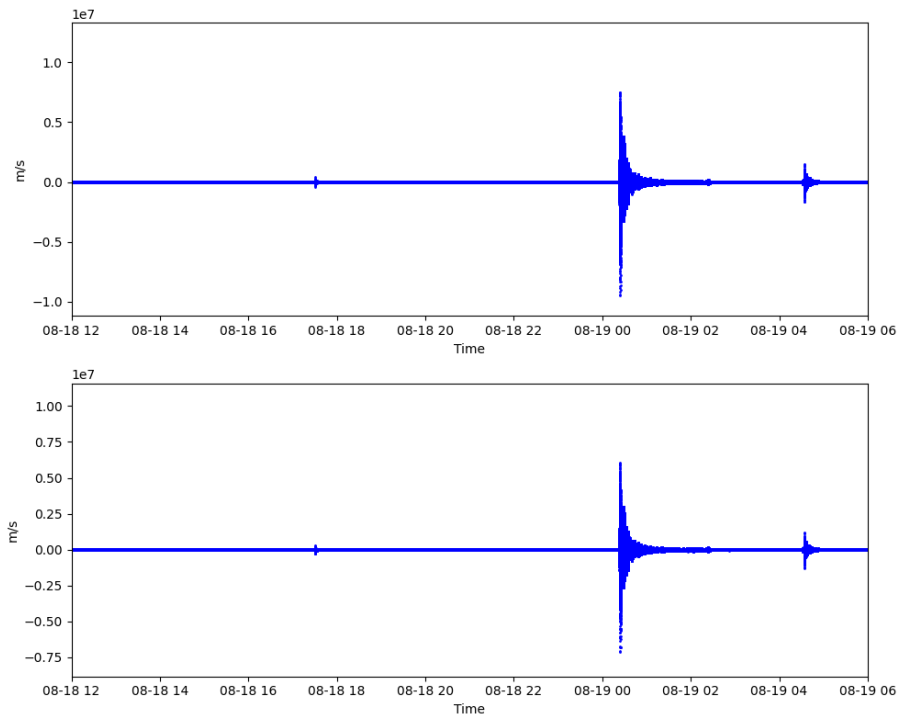


Figure 11: Raw Fiji Earthquake Time Series Data from RAO station